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THEORETICAL/REVIEW

A meta-analytic review of gaze cueing across the autism trait continuum

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Abstract
Social attention is fundamental to social interaction and is experimentally indexed by the gaze-cueing effect (GCE) derived from the gaze-cueing task. Although atypical social attention is often considered a core feature of autism spectrum disorder (ASD), empirical findings regarding the GCE in ASD remain heterogeneous. This inconsistency highlights the need to investigate social attention across the broader autism trait continuum (autism-like traits [ALT]) in the general population to clarify these mechanisms. This meta-analysis synthesized 28 correlational studies (77 effect sizes) and eight group-comparison studies (21 effect sizes) to resolve these inconsistencies. A three-level model integrated the effect sizes, with moderation analyses testing the influence of sample characteristics (proportion of males, sample size), ALT measure type, and experimental parameters (face type, expression, SOA, task type). Results revealed a significant, albeit small, negative correlation: higher ALT levels predicted reduced GCE. Conversely, group comparisons (high ALT vs. low ALT) yielded nonsignificant results. These findings align with the continuum hypothesis, suggesting a gradient relationship rather than a distinct subgroup effect, although statistical power in group comparisons should also be considered. Moderation analyses indicated that the negative association intensified in samples with a higher male proportion but weaker in those with larger sample sizes; no other moderators were significant. Together, these findings indicate that the link between autistic traits and social attention is subtle, continuous, and moderated by sample characteristics, underscoring the necessity for future research using more ecologically valid paradigms to capture the dynamic nature of social processing.

Keywords Social attention · Autism-like traits · Social cognition · Meta-analysis

Introduction

During social interactions, individuals rely on social cues such as head orientation, gestures, and gaze direction to infer others' intentions and mental states, and to predict subsequent behavior (Cañigueral & Hamilton, 2019; Frith & Frith, 2007; Ristic & Capozzi, 2022). Among these cues, averted gaze plays a critical role in triggering attentional orienting, a phenomenon foundational to social cognition and theory of mind (ToM; Cañigueral & Hamilton, 2019; Capozzi & Kingstone, 2024; Frischen et al., 2007; Ristic & Capozzi, 2022). Although atypical social attention is widely considered as a characteristic feature of autism spectrum disorder (ASD; American Psychiatric Association, 2013), its expression varies substantially across individuals. Research indicates that individuals with ASD exhibit reduced sensitivity to social cues and altered prioritization of social versus nonsocial information (Chevallier et al., 2012; Frazier et al., 2017; Guillon et al., 2014; Nakano et al., 2010; Vivanti et al., 2017). Such atypicalities may contribute to core challenges in social interaction and communication (Frischen et al., 2007; Mundy, 2023). Consequently, social attention patterns have been proposed as both early behavioral markers and potential diagnostic indicators of ASD (Kang et al., 2018; Liu et al., 2021). Thus, characterizing social attention

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in ASD is crucial for understanding the mechanisms underlying their unique social functioning deficits.

To investigate the mechanisms underlying social attention in ASD, researchers commonly employ the gaze-cueing task, a modification of the Posner cueing paradigm (Capozzi & Ristic, 2018; Posner, 1980). In this task, a central face cue (gazing left or right) precedes a target stimulus. Typically, participants respond faster to targets appearing at the cued location (congruent trials) relative to the uncued location (incongruent trials), a phenomenon known as the gaze-cueing effect (GCE; Dalmaso et al., 2020; Dodd et al., 2011; Friesen & Kingstone, 1998; Frischen et al., 2007). Studies utilizing the gaze-cueing task in ASD populations have yielded inconsistent results. While some studies report reduced GCEs or delayed attentional orienting in ASD (Liu et al., 2021; Vivanti et al., 2017), others observe intact performance, particularly in high-functioning individuals or under specific conditions (Ristic et al., 2005). These inconsistencies have been attributed to sample heterogeneity (e.g., intellectual ability) and methodological variability. Several theoretical frameworks address the nature of these social attention deficits: the social motivation hypothesis posits a reduced intrinsic drive to process social signals (Chevallier et al., 2012; Guillon et al., 2014; Liu et al., 2021; Müeller & Fishman, 2018); the social cognition hypothesis suggests difficulties in extracting mentalistic information from gaze (Mundy & Bullen, 2022); and the threat avoidance hypothesis argues that eye contact elicits aversive arousal, prompting avoidance behavior (Cuve et al., 2018; Hadjikhani & Johnels, 2023; He et al., 2019; Stuart et al., 2023). Crucially, these theories imply a mechanism extending beyond clinical diagnosis: If these cognitive and motivational processes are continuous, variations in ALT within the general population should similarly predict individual differences in social attention.

To investigate this continuity, recent research has focused on autism-like traits (ALT) in nonclinical populations to clarify these associations. ALT refers to subclinical autistic characteristics that are continuously distributed across the general population, with diagnosed ASD representing the extreme upper end of this spectrum (Hoekstra et al., 2007; Zhang et al., 2016; Zhang et al., 2020). Investigating ALT offers a key advantage over clinical group comparisons by allowing researchers to examine the dimensional nature of social attention without the confounding effects of medication, intellectual disability, or comorbidities often present in clinical samples (Jones et al., 2013). However, empirical findings regarding the association between ALT and gaze cueing remain inconclusive. Early studies suggested a negative correlation, wherein higher ALT (e.g., measured by the Autism-Spectrum Quotient [AQ]) predicted weaker GCEs (Bayliss & Tipper, 2005; Bayliss et al., 2005; Edwards et al., 2015). However, subsequent replications have been inconsistent. Some studies failed to detect any significant association (Edwards et al., 2015), while others even reported positive correlations (Dalmaso et al., 2016) or context-dependent effects (e.g., only under low-predictability conditions; Ristic et al., 2005). Furthermore, behavioral patterns in individuals with high-ALT may not simply mirror atypicality observed in ASD. For instance, some individuals exhibit heightened sensitivity to social cues, potentially driven by compensatory social motivation or loneliness (Chevallier et al., 2012; Pickett et al., 2004). These divergent findings imply a complex relationship between ALT and social attention, likely modulated by underlying cognitive mechanisms (e.g., automatic vs. controlled processing; Lin et al., 2020) or methodological factors.

Notably, the observed heterogeneity in findings may stem from variations in experimental paradigms and trait conceptualizations. Regarding experimental parameters, GCE magnitude is modulated by the ecological validity of the cue (e.g., schematic vs. real faces; Dalmaso et al., 2020; McKay et al., 2021), expression (e.g., fearful faces typically elicit stronger cueing; McKay et al., 2021), stimulus onset asynchrony (SOA; Frischen et al., 2007), and task type (e.g., detection vs. localization vs. categorization; Chacón-Candia et al., 2023; McKay et al., 2021). It remains unclear whether ALT-related attentional differences are robust across these conditions or restricted to specific conditions. Furthermore, methodological diversity in defining the multidimensional ALT construct poses a significant challenge for synthesis. Standardized tools like the AQ and Social Responsiveness Scale (SRS) are used to directly quantify these traits (Bayliss & Tipper, 2005; Uono et al., 2022). However, researchers also frequently investigate constructs theoretically linked to the autism phenotype, such as empathy (Alwall et al., 2010; Shin et al., 2023), ToM (Baron-Cohen, 2009; Baron-Cohen & Wheelwright, 2004), and social anxiety (Edwards & Bayliss, 2019). These related traits are often employed as proxies or component processes of the broader autism phenotype. To ensure a comprehensive synthesis, we adopted an inclusive approach incorporating both direct trait measures and assessments of overlapping social-cognitive constructs. Furthermore, we explicitly examined the moderating role of measurement type to test effect specificity.

Although early studies identified atypical social attention in individuals with high ALT (Bayliss & Tipper, 2005; Bayliss et al., 2005; Edwards et al., 2015), subsequent findings have been fragmented and inconsistent. To date, no systematic synthesis has quantitatively evaluated this relationship in the general population. While meta-analyses exist regarding attentional orienting in clinical ASD (Murza et al., 2016; Todorova et al., 2019) or general gaze-cueing effects (Chacón-Candia et al., 2023; McKay et al., 2021), the specific association between ALT and social attention remains unexamined. Examining how ALT variation relates to gaze-cueing performance holds significant theoretical and

practical implications for understanding the mechanisms underlying ASD (Wang et al., 2020). The present meta-analysis aims to fill this gap by synthesizing correlational and group-comparison studies to determine the magnitude and robustness of the association between ALT and the GCE. Specifically, we sought to (1) quantify the overall strength of the relationship to test the continuum hypothesis of social attention; (2) examine the influence of sample characteristics (e.g., sex ratio, sample size); and (3) systematically test whether the effect is moderated by ALT measure type (e.g., explicit ALT vs. related social-cognitive constructs) or experimental parameters (e.g., face type, expression, SOA, task type). This study seeks to provide empirical evidence clarifying social attention mechanisms across the autism trait continuum and to resolve existing inconsistencies regarding the ALT–social attention relationship.

Materials and methods

Search strategy and inclusion criteria

We conducted a comprehensive literature search across the following electronic databases: Web of Science, Scopus, PubMed, PsycINFO (via EBSCOhost), and MEDLINE. The search was restricted to studies with titles or abstracts containing the following keywords: ("autism-like traits" OR "autism traits" OR "autistic traits" OR "autistic-like traits" OR "AQ" OR "Autism-Spectrum Quotient" OR "Autism quotient") AND ("social attention" OR "joint attention" OR "shift attention" OR "attentional shifts" OR "attentional orienting" OR "social orienting" OR "shared attention" OR "gaze following" OR "following gaze" OR "gaze shifts"). The search was restricted to articles published in English.

The initial search was completed by the second week of November 2024, yielding 4,567 records. Following the removal of 661 duplicates, 3,906 unique citations (spanning from 1967 to 2025, including early-access publications) were retained. To ensure the review remained current, an updated search was conducted in December 2025, covering the period from 2024 to the present. This update yielded an additional 120 records. After excluding 57 duplicates (10 overlapping with the initial search and 47 new duplicates), 63 unique records were added for screening. Both abstracts and full texts were screened according to the following inclusion and exclusion criteria:

1 Only peer-reviewed empirical studies were included; review articles, opinion pieces, and case studies were excluded.
2 Participants were typically developing individuals, with their ALT assessed using validated questionnaires. Eligible measures included standardized instruments (e.g., AQ, SRS) or validated proxies for social-cognitive constructs (e.g., empathy, social anxiety, ToM). Proxy measures were included only if explicitly conceptualized by the original authors as operationalizations of the broader autism phenotype.
3 Studies must employ the gaze-cueing task, which includes both cue and target stimuli. Following the presentation of a gaze cue, a target appears, and participants are required to respond. Face types could include schematic, computer-generated, or real faces. Both central and peripheral cueing tasks were eligible, though the analysis primarily focused on central gaze cues.
4 Task types included detection, localization, and categorization: detection tasks required participants to detect the appearance of a target; localization tasks required identifying the target's spatial location; and categorization tasks required classification of the target.
5 Participants must not have undergone any prior interventions that could modulate the properties of the gaze cue or social attention mechanisms before performing the task (e.g., oxytocin administration, social exclusion priming, or manipulation of cue trustworthiness).
6 The study must examine the association between social attention and ALT, typically assessed via the GCE. The GCE was defined as the behavioral performance difference between congruent and incongruent trials, primarily quantified as reaction time differences (RT_incongruent – RT_congruent), although accuracy rates or inverse efficiency scores were also accepted if reported as the primary outcome.
7 Only studies reporting correlational analyses or group comparisons (e.g., between low and high ALT groups based on questionnaire scores) were included.

Study selection

The study selection process followed PRISMA guidelines, as illustrated in Fig. 1. The first and second authors independently screened the titles and abstracts of all retrieved studies to assess their eligibility. If both reviewers agreed to exclude a study, it was removed. If both agreed to include a study, it was retained for full-text review. Disagreements were resolved through discussion or consultation with the corresponding author.

Across both the initial and updated searches, a total of 3,969 unique records (3,906 from initial, 63 from updated) were screened at the abstract stage. Of these, 3,636 records were excluded (3,584 from initial, 52 from updated). The remaining 333 articles (322 from initial, 11 from updated) underwent full-text screening.

Full-text screening led to the exclusion of 299 studies (288 from initial, 11 from updated). Reasons for exclusion included: conference abstracts (four from initial), dissertations (six from

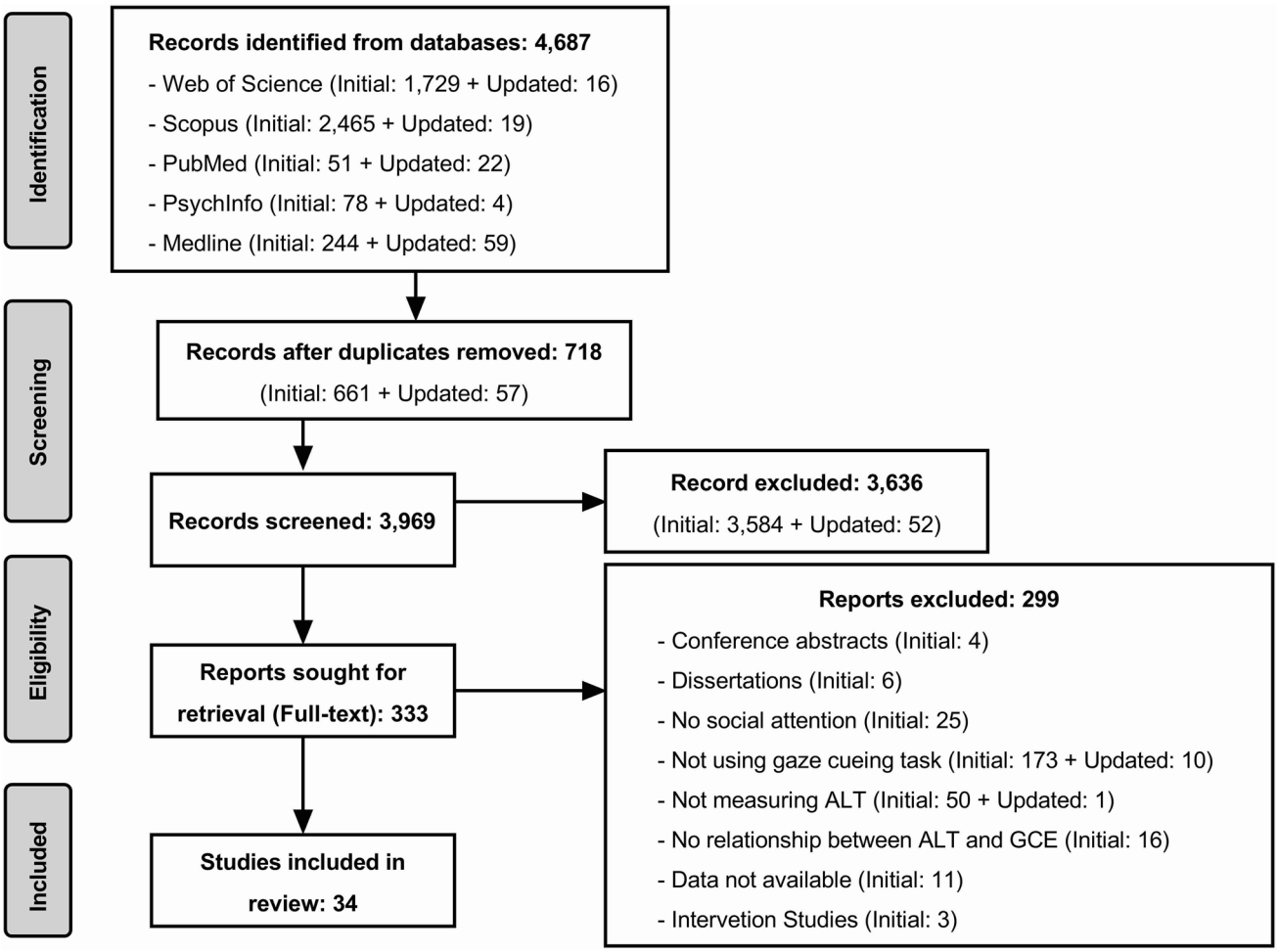

Fig. 1 PRISMA flowchart illustrating the study screening and selection process. The reported numbers integrate records from both the initial search (November 2024) and the updated search (December 2025)

initial), not examining social attention (25 from initial), not employing a gaze-cueing task (173 from initial, 10 from updated), not measuring ALT or related characteristics in TD individuals (50 from initial, one from updated), not examining the relationship between ALT and cueing effects (16 from initial), data note available (11 from initial), and intervention studies (three from initial). Ultimately, 34 studies met all inclusion criteria. All included studies employed a gaze-cueing task and reported either correlational analyses of ALT and cueing effects or group comparisons based on ALT levels.

Study coding and data extraction

The first and second authors independently coded all 34 included studies. The first author cross-checked the second author's coding results for accuracy. The following study-level variables were extracted (see Appendix A in Supplementary Materials for the full coding table):

1 Demographic characteristics, such as initial sample size, mean age, and proportion of males.
2 ALT measure type for correlational studies: Measures were stratified into three distinct categories: (1) Explicit Autistic Traits, comprising the Social Responsiveness Scale (SRS), Broad Autism Phenotype Questionnaire (BAPQ), and variants of the Autism-Spectrum Quotient (AQ; including the AQ-10 and both binary- and Likert-scored versions of the AQ-50); (2) Related Social-Cognitive Constructs, including measures of Empathy (Empathy Quotient [EQ], Questionnaire of Cognitive and Affective Empathy[QCAE]), Social Anxiety (Social Phobia Inventory [SPI], Liebowitz Social Anxiety Scale [LSAS], Social Interaction Anxiety Scale [SIAS]), and Theory of Mind (Yoni task, Reading the Mind in the Eyes [RME] test); and (3) Other Associated Traits, such as the Systemizing Quotient (SQ) and UCLA Loneliness Scale.

3 ALT measure type for group comparison studies: Explicit Autistic Traits, measured using the AQ-50, AQ-200, BAPQ, or clinical diagnostic criteria.
4 Face type is categorized as schematic, computer-generated, or real faces.
5 Expression: Angry, Fearful, Happy, Surprised, Neutral, and Mixed. The "Mixed" condition means that multiple expression conditions were used in the study, but their associations were not analyzed across expressions.
6 SOA is categorized as Short (SOA ≤ 300 ms), Middle (300 ms < SOA ≤ 600 ms), Long (600 ms < SOA), and Mixed. The "Mixed" condition indicates that multiple SOA conditions were used in the study, but their associations were not analyzed across SOA.
7 Task type: Detection, Localization, or Categorization.

Data extraction was restricted to results using upright cue presentations. Inverted cues were excluded as they disrupt holistic face processing and typically elicit qualitatively different orienting mechanisms (Jenkins & Langton, 2003; Schwaninger et al., 2005). Furthermore, only a negligible number of effect sizes ($k = 2$ from a single study) were available for this condition, precluding meaningful synthesis.

For correlational studies, Pearson's correlation coefficients and standardized regression coefficients (β) were extracted and transformed to Fisher's z values. Effect signs for measures of social competence (e.g., EQ, QCAE, ToM tasks) were reversed to ensure that all effect sizes consistently reflected the association between higher autism-like traits (or lower social competence) and gaze cueing. Consequently, a negative Fisher's z indicates that higher ALT is associated with diminished GCE magnitudes.

For group comparison studies, means, standard deviations, sample sizes, and F values for the low and high ALT groups were extracted to calculate Hedges' g. A positive Hedges's g indicates that the high-ALT group exhibits a smaller difference between congruent and incongruent trials (i.e., a weaker GCE) compared to the low-ALT group. If studies reported only the means and standard deviations for the congruent and incongruent conditions, rather than direct cueing effects for each group, the standard deviation of the cueing effect was calculated using the following equation. Referring to previous studies, we set r to 0.70 (Chacón-Candia et al., 2023):

$$SD_{GCE} = \sqrt{SD_{invalid} + SD_{valid} - 2 \times r \times SD_{invalid} \times SD_{valid}}$$

Quality assessment

The first and second authors independently assessed the risk of bias for all 34 studies using the Standard Quality Assessment (SQA) tool (Kmet et al., 2004). This tool evaluates methodological quality across domains, including study design, subject selection, outcome measurement, and analysis. The original 14-item checklist was adapted for the present review: items related to randomization and blinding in intervention studies (Q5, Q6, Q7) were excluded as they did not apply to the correlational and group-comparison designs included in this analysis. The remaining items assessed criteria such as whether the research objective was sufficiently described (Q1), whether outcome measures were robust to bias (Q8), and whether appropriate statistical controls for confounding were employed (Q12). Item was scored as "Yes" (2 points), "Partial" (1 point), "No" (0 points), or "N/A." A summary quality score was computed for each study using the following formula:

$$Total\ sum = \left(number\ of\ ''\text{Yes}'' \times 2\right) + \left(number\ of\ ''paritals'' \times 1\right)$$

$$Total\ possible\ sum = 22 - \left(number\ of\ ''\text{N/A}'' \times 2\right)$$

The summary quality score was calculated as the total sum divided by the total possible sum. To ensure reliability, both authors randomly selected 10 studies for initial quality assessment and compared their total scores to verify that the difference did not exceed three points. If the difference exceeded three points, a final score was determined through discussion. Subsequently, the remaining 24 studies were assessed for quality.

The Spearman correlation coefficient between the total assessment scores of the two evaluators was 0.568, $p < .001$. The final score for each study was the average of the two evaluators' scores. The overall score was 0.90 ± 0.05, with a range of 0.80 to 1.00.

Statistical analysis

All data analyses were performed using the R package *metafor*. The *metafor* package utilizes Fisher's z to measure the correlation effect and Hedges's g to assess the effect size of group differences. Specifically, a negative Fisher's z indicates a negative correlation, meaning that as ALT scores increase, the magnitude of the GCE decreases (i.e., weaker social attention). A positive Hedges' g suggests that the high ALT group exhibits a smaller difference in attentional orienting between congruent and incongruent trials compared to the low ALT group.

Given that most studies reported multiple results (e.g., analyses with different emotional faces as cue stimuli or different SOAs) derived from the same sample, we conducted a three-level meta-analysis using the "rma.mv" function in the *metafor* package. Unlike a traditional two-level model, which assumes independence between effect sizes, a three-level model explicitly models variance at three distinct levels: sampling variance of the observed

effect sizes (Level 1), heterogeneity between effect sizes nested within the same study (Level 2, within-study variance), and heterogeneity between different studies (Level 3, between-study variance). This approach minimizes data loss and enhances statistical power while controlling for the dependency of effect sizes. Model fit was compared against a standard two-level model using analyses of variance (ANOVAs) and information criteria (AIC/BIC). The "var.comp" function in the *dmetar* package was used to calculate the variance distribution across these levels. Heterogeneity was assessed using Cochran's Q-statistic (reported as QE) and the I^2 index, which quantifies the proportion of total variance attributable to heterogeneity rather than sampling error.

Moderator analyses were conducted using the rma.mv function in the *metafor* package. Tested moderators included sample characteristics (proportion of males, sample size), ALT measure type, and experimental parameters (face type, expression, SOA, task type). Moderator selection was guided by theoretical relevance (see Introduction) and data availability (see Appendix A in Supplementary Materials).

Finally, we assessed the risk of publication bias using visual inspection of funnel plots and Egger's regression test to quantify asymmetry. If significant bias was detected, we employed the trim and fill method (using a standard random-effects model) to estimate missing studies and to calculate an adjusted overall effect size. This sensitivity analysis evaluated the robustness of the observed effects against small-study effects or selective reporting bias.

Results

Sample description

As shown in Fig. 1, this study included a total of 34 studies published between 2005 and 2024. Among these, 26 studies reported correlation coefficients, six studies examined group differences, and two studies included both correlation and group difference analyses.

The correlational studies involved 77 effect sizes, with a total of 2,161 participants recruited, and an average age of 21.53 years. Among these, 800 were male, accounting for 37.03%. The geographic distribution of the included studies was as follows: Asia ($n=8$ studies), Australia ($n=1$), Europe ($n=15$), and North America ($n=4$).

The group difference studies involved 21 effect sizes, with a total of 464 participants and an average age of 23.17 years. Among these, 136 were male, accounting for 29.31%. The included studies were distributed across Asia ($n=6$) and Europe ($n=2$).

Overall effect in correlational studies

Main analysis

The three-level random-effects meta-analysis, including all ALT measure types ($k=77$ effect sizes), revealed a significant negative correlation. The average effect size was $z=-0.10$ ($SE=0.03$), $t(76)=-3.83$, $p<.001$, 95% CI [−0.15, −0.05]. Converting this to Pearson's r yields −0.10, indicating a weak but significant negative association: higher ALT scores predict reduced GCE magnitudes. Significant heterogeneity was detected between studies, QE(76) = 153.61, $p<.001$.

To validate the three-level model, a standard two-level model was fitted for comparison. The results showed that the average estimated effect size was $z=-0.10$ ($SE=0.02$), $t(76)=-5.18$, $p<.001$, 95% CI [−0.14, −0.06]. The three-level model demonstrated superior performance (AIC = −45.54, BIC = −38.54, AICc = −45.20, logLik = 25.77) compared with the two-level model (AIC = −42.88, BIC = −38.21, AICc = −42.71, logLik = 23.44). Additionally, the likelihood ratio test (LRT) results showed $\chi^2(1)=4.66$, $p=.031$, indicating significantly better model fit. Consequently, the three-level structure was retained for subsequent analyses.

Variance distribution for the three-level model was estimated using I^2, which reflects the proportion of total variance attributable to within-study and between-study variance. This calculation was performed using the "var.comp" function in the *dmetar* package. The results indicated that 49.36% of the variance was attributable to heterogeneity, within-study variance accounted for 19.04%, and between-study variance for 30.33%, while sampling error accounted for the remaining 50.64%. As the sampling variance fell below the 75% threshold (Assinka & Wibbelink, 2016), the observed heterogeneity was considered substantial. Accordingly, we examined the potential moderating effects of study characteristics (e.g., sample characteristics, ALT measure type, and experimental parameters). The forest plot shown in Fig. 2a displays the effect sizes and their confidence intervals clustered by all ALT measure types.

Subgroup analysis: Explicit autistic traits

To evaluate effect specificity and address measurement heterogeneity, we conducted a sensitivity analysis restricted to studies using "explicit autistic traits" measures (e.g., AQ, SRS, BAPQ; $k=46$ effect sizes). The results revealed a robust and slightly stronger negative correlation ($z=-0.12$, $SE=0.03$), $t(45)=-4.75$, $p<.001$, 95% CI [−0.18, −0.07]; equivalent to $r=-0.13$). Significant heterogeneity persisted in this subgroup, QE(45) = 84.53, $p<.001$. Forest plots for the entire sample and the subgroup are shown in Fig. 2b.

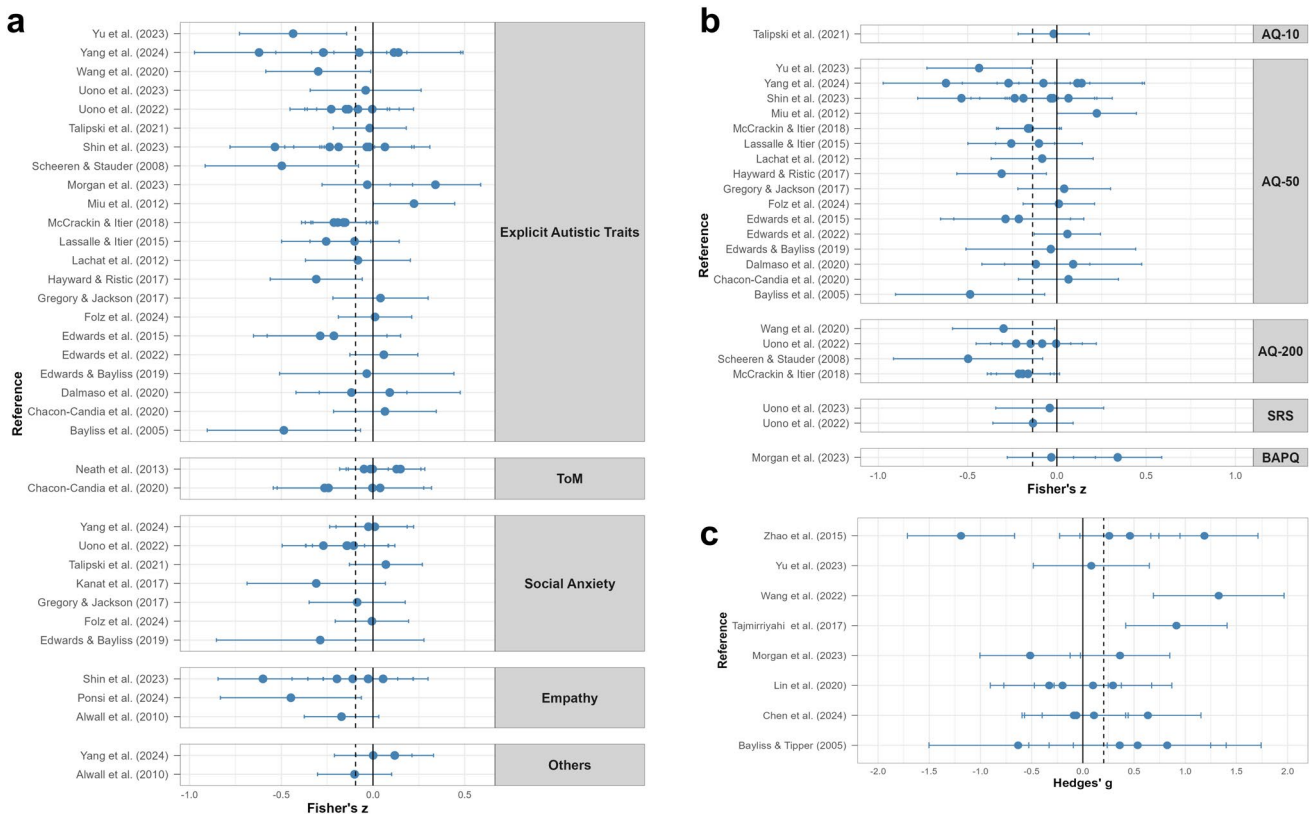

Fig. 2 Forest plots showing the effect sizes (Fisher's z and Hedges's g) and corresponding confidence intervals for each study. **a** Forest plot for correlational studies across all ALT measure types. **b** Forest plot for correlational studies restricted to the "Explicit Autistic Traits" subgroup. **c** Forest plot for group-comparison studies

Overall effect in group comparison studies

The three-level analysis of the group differences study revealed an average effect size of Hedges's $g = 0.20$ ($SE = 0.13$), $t(20) = 1.54$, $p = .140$, 95% CI [−0.07, 0.48]. Although the direction of the effect (positive g) is consistent with the hypothesis that high-ALT groups show weaker cueing effects than low-ALT groups, this difference was not statistically significant. Significant heterogeneity was observed, QE(20) = 87.00, $p < .001$. The forest plot is displayed in Fig. 2c. Moderator analyses were omitted for this subgroup due to the non-significant main effect and insufficient sample size ($n < 10$). The lack of statistical significance may stem from reduced statistical power associated with small sample sizes or from information loss inherent in dichotomizing continuous traits.

Moderator analyses in correlational studies

We conducted separate univariate meta-regression analyses for continuous moderators (proportion of males, sample size) and subgroup analyses for categorical moderators (ALT measure type, face type, expression, SOA, task type).

Sample characteristics

Proportion of males: The mean proportion of males was 0.37, ranging from 0.00 to 1.00. Moderation analysis revealed a significant effect, $F(1, 75) = 5.14$, $p = .026$. A negative regression coefficient ($\beta = -0.33$, $SE = 0.15$) indicated that a higher male proportion was associated with a stronger (more negative) correlation between ALT and GCE (Fig. 3a). Residual heterogeneity remained significant, QE(75) = 148.73, $p < .001$.

Sample size: With a mean sample size of 63.86, ranging from 15 to 222. Sample size significantly moderated the effect, $F(1, 75) = 6.65$, $p = .011$. The positive coefficient ($\beta = 0.001$, $SE = 0.001$) suggests that the negative association diminished (approached zero) in larger samples (Fig. 3b). Residual heterogeneity remained significant, QE(75) = 130.81, $p < .001$.

ALT measure type

To assess the effect of specificity, ALT measure type was included as a moderator, with explicit autistic traits set as the reference category. The moderation test was nonsignificant,

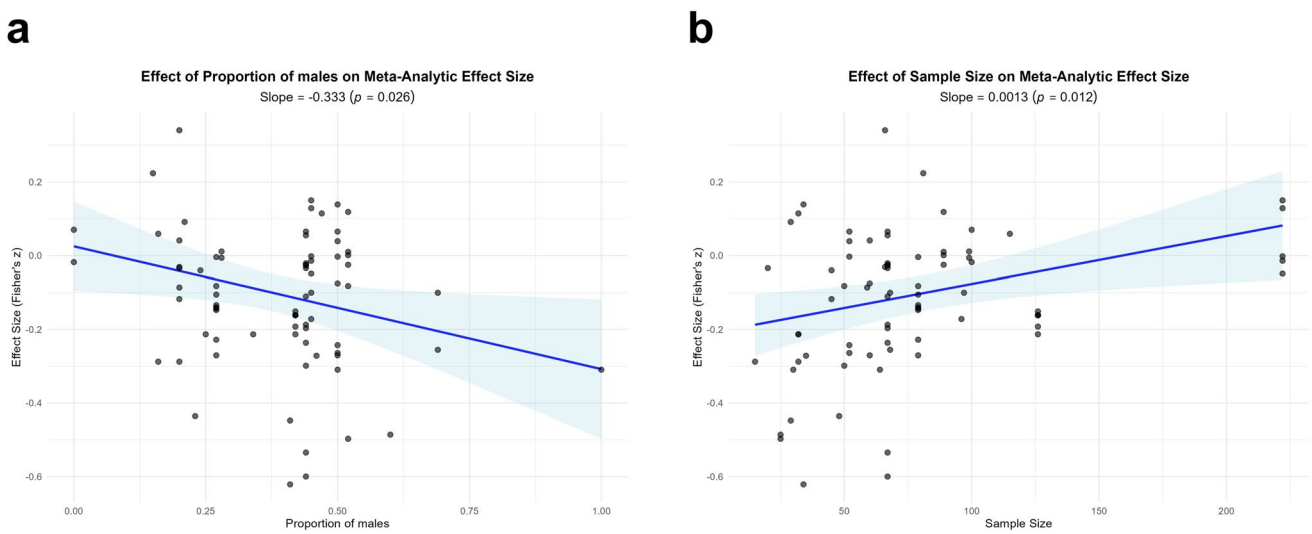

Fig. 3 Meta-regression of proportion of males (**a**) and sample size (**b**) on effect size. The solid blue line represents the predicted effect sizes (REML estimation). The shaded region indicates the 95% confidence interval around the regression slope. Individual data points correspond to effect sizes from included studies

$F(4, 72) = 1.00$, $p = .411$, suggesting that the association strength did not statistically differ across the various constructs used to operationalize the phenotype.

Specifically, under the Explicit Autistic traits subgroup, the average effect size was significantly different from zero ($z = -0.12$), $t(72) = -4.16$, $p < .001$, confirming a robust negative association. Compared with this reference group, did not differ significantly for empathy ($z = -0.16$), $t(72) = -0.73$, $p = .471$, social anxiety ($z = -0.09$), $t(72) = 0.72$, $p = .719$, ToM ($z = -0.02$), $t(72) = 1.44$, $p = .154$, or other ($z = -0.01$), $t(72) = 1.13$, $p = .264$, categories.

As shown in Fig. 4a, although the negative correlation appeared numerically strongest for Empathy category and weakest for ToM/Other categories, these differences were not statistically reliable. Crucially, the finding that the explicit autistic traits subgroup showed a significant negative effect independently of related constructs reinforces the validity of the main findings in the Subgroup Analysis section.

Experimental parameters

We assessed the moderating role of four experimental parameters. None yielded a significant overall moderation effect.

Face type The type of face cue did not significantly moderate the effect, $F(2, 74) = 0.64$, $p = .532$. Using computer-generated faces which was significantly different from zero as the reference ($z = -0.14$), $t(74) = -2.31$, $p = .024$, effect sizes for real faces ($z = -0.08$), $t(74) = 0.85$, $p = .398$, and schematic faces ($z = -0.16$), $t(74) = -0.17$, $p = .864$, did not differ significantly. This suggests that the ecological validity of the cue does not strongly influence the ALT-GCE relationship (Fig. 4b).

Expression Expression also showed no significant moderation, $F(5, 71) = 1.11$, $p = .365$. With angry faces ($z = -0.06$) as the reference, effect sizes for fearful ($z = -0.09$), $t(71) = -0.32$, $p = .754$, happy ($z = -0.08$), $t(71) = -0.22$, $p = .824$, surprised ($z = -0.10$), $t(71) = -0.34$, $p = .739$, neutral ($z = -0.14$), $t(71) = -0.92$, $p = .362$, and mixed ($z = 0.02$), $t(71) = 0.78$, $p = .439$, did not differ significantly (Fig. 4c).

SOA SOA did not significantly moderate the effect, $F(3, 73) = 1.27$, $p = .290$. Although the middle SOA condition (300–600 ms) showed a numerically larger negative effect ($z = -0.18$) compared to the reference long SOA ($z = -0.09$), this difference approached but did not reach significance, $t(73) = -1.30$, $p = .198$. Short ($z = -0.08$), $t(73) = 0.22$, $p = .827$, and mixed ($z = -0.05$), $t(73) = 0.34$, $p = .734$, SOAs also did not differ from the reference (Fig. 4d). It is worth noting that while the Long SOA baseline was not significantly different from zero, the Middle SOA effect size was numerically the most robust.

Task type Finally, the task requirement did not influence the results, $F(2, 74) = 0.10$, $p = .908$. Effect sizes were virtually identical across categorization (reference, $z = -0.10$), different from zero, $t(74) = -2.33$, $p = .022$, detection ($z = -0.08$), $t(74) = 0.29$, $p = .772$, and localization ($z = -0.11$), $t(74) = -0.19$, $p = .847$, tasks (Fig. 4e).

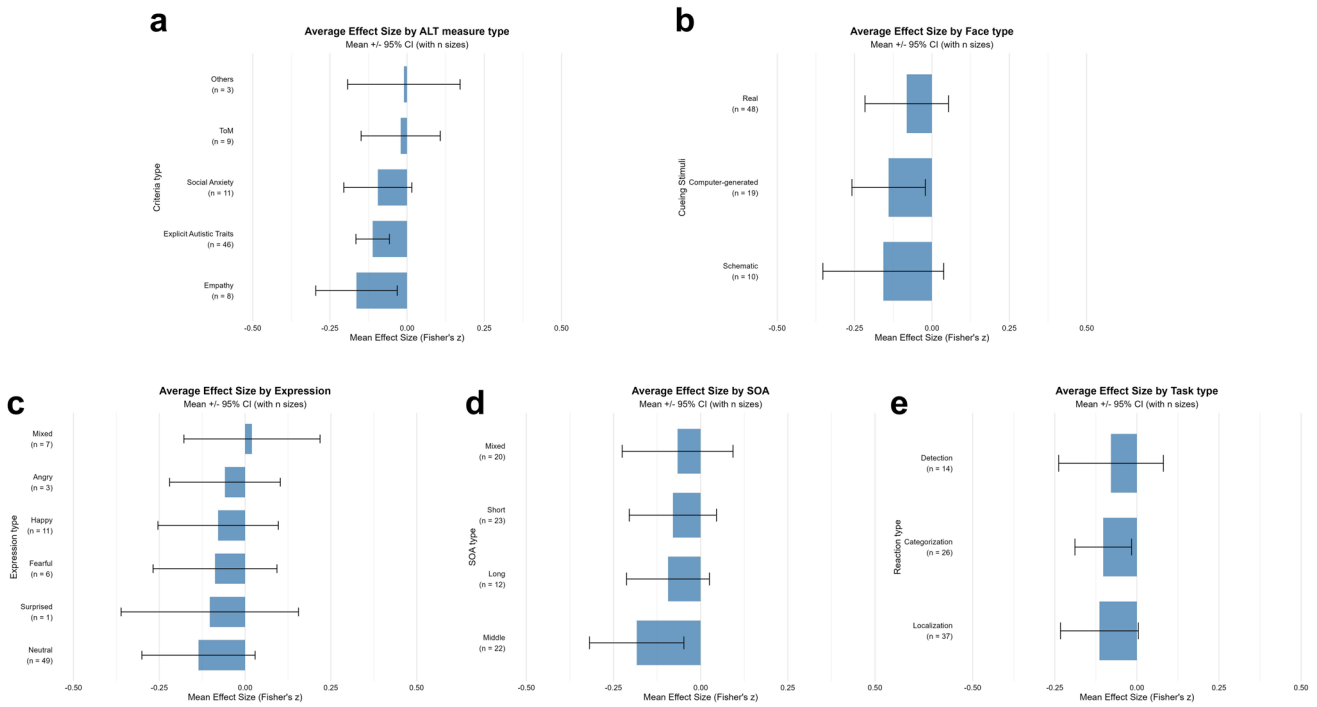

Fig. 4 Horizontal bar plots comparing mean effect sizes with 95% confidence intervals across ALT measure type (**a**), Face type (**b**), Expression (**c**), SOA (**d**), and Task type (**e**)

Publication bias

First, visual inspection of the funnel plot suggested approximate symmetry, though slight asymmetry was observed on the right side. Egger's regression test using a random-effects model confirmed significant publication bias, $t(75) = -3.30$, $p = .002$.

Next, we applied the Trim and Fill method using a standard random-effects model. As shown in Fig. 5, the analysis imputed two missing studies on the right side of the funnel

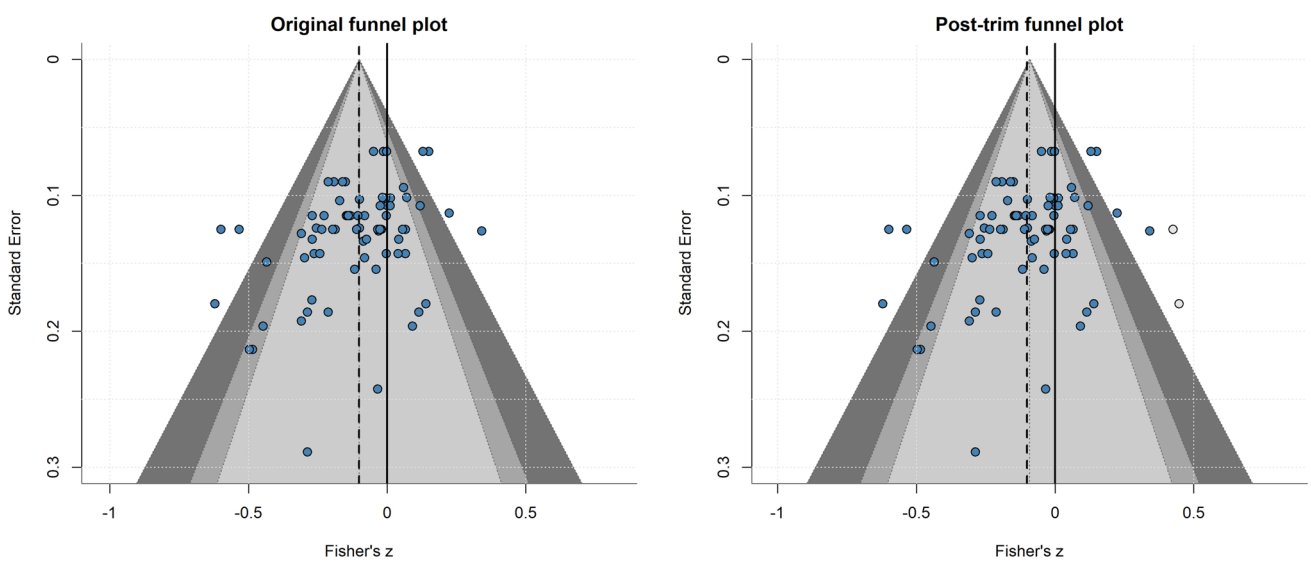

Fig. 5 Funnel plots displaying the original data (left) and the trim-and-fill adjusted results (right). The dashed vertical line indicates the overall effect size. Blue circles represent individual effect sizes from correlational studies, while white circles indicate imputed effect sizes added by the trim-and-fill procedure to address potential publication bias. (Color figure online)

plot. After adjustment, the average effect size remained virtually unchanged and statistically significant ($z = -0.09$, $SE = 0.02$), $t(78) = -4.41$, $p < .001$, 95% CI [−0.13, −0.05]. This suggests that while there is evidence of selective reporting, it does not fundamentally alter the observed negative association between ALT and GCE.

Discussion

This meta-analysis synthesized existing evidence regarding the relationship between ALT and social attention (indexed by the gaze-cueing effect, GCE) in the general population, effectively testing the continuum hypothesis of autistic characteristics (Bayliss & Tipper, 2005; Bayliss et al., 2005; Edwards et al., 2015). Integrating data from 34 studies, we identified a significant, though weak, negative correlation: individuals with higher ALT exhibited reduced social attentional orienting. Crucially, this association remained robust and was numerically stronger when restricted to studies using explicit autistic trait measures, supporting the specificity of the link.

Moderator analyses indicated that this relationship was significantly influenced by sample characteristics. Specifically, the negative correlation was stronger (more negative) in samples with a higher proportion of males and weaker (diminished) in studies with larger sample sizes. In contrast, experimental parameters (e.g., face type, expression, SOA, task type) and ALT measure type did not significantly moderate the effect, suggesting an invariant influence of ALT across different contexts. Group comparison studies yielded a non-significant trend in the expected direction (high-ALT < low-ALT), likely due to limited statistical power.

Sample characteristics

First, the proportion of males emerged as a significant moderator: as the proportion of males increased, the negative correlation between ALT and cueing effects intensified. This finding aligns with the "extreme male brain" theory and the observation that males typically exhibit higher baseline ALT scores (Baron-Cohen, 2002). A plausible explanation is that samples with a higher male ratio capture a broader segment of the ALT spectrum, thereby increasing the statistical variance necessary to detect a correlation. Conversely, samples with fewer males may suffer from range restriction in ALT scores, attenuating the observed association. Additionally, males may be more efficient at inhibiting reflexive orienting to nonpredictive social cues compared with between high ALT (associated with male-typical cognition) and reduced gaze cueing.

Second, sample size significantly moderated the effect: The negative correlation was most pronounced in smaller samples and diminished (approaching zero) in larger studies. This pattern suggests the presence of small-study effects or publication bias, wherein smaller studies with large effect sizes are more likely to be published (Ponsi et al., 2024; Scheeren & Stauder, 2008), while larger studies often report null or attenuated results (Edwards et al., 2022; Folz et al., 2024; Neath et al., 2013). Alternatively, it may suggest that the true effect size of the ALT-GCE association in the general population is inherently subtle. If the effect is weak ($r \approx -0.09$ or smaller), small samples lack the statistical power to reliably estimate it, leading to a "Winner's Curse" phenomenon where only exaggerated estimates reach significance. This finding underscores the necessity for future research to employ sufficiently powered designs (based on a priori power analysis for small-to-medium effects) to accurately capture this nuanced relationship. Furthermore, given the subtle nature of the effect, correlational approaches utilizing the full range of trait variability offer greater statistical efficiency than group-based designs, which suffer from information loss due to dichotomization. Indeed, our publication bias analysis confirmed significant asymmetry, and the Trim and Fill method identified missing studies. However, the overall negative correlation remained significant after adjustment, suggesting that while selective reporting may inflate estimates, it does not fully account for the observed association.

Measurement specificity for ALT

A critical challenge in synthesizing the ALT literature is the methodological diversity in defining the phenotype. While some studies employ explicit measures like the AQ and SRS (Bayliss et al., 2005; Uono et al., 2023), others rely on related constructs such as empathy or social anxiety as proxies for the broader autism phenotype (Barbeau et al., 2009; Baron-Cohen, 1995; Chevallier et al., 2012; Morgan et al., 2023). Our moderation analysis indicated that the type of measure used did not significantly moderate the overall effect size, suggesting a degree of convergence across these diverse conceptualizations.

However, specificity is key. When analysis was restricted to explicit autistic trait measures (e.g., AQ, SRS), the negative correlation with gaze cueing remained robust ($r = -0.13$) relative to the full sample ($r = -0.09$). While these overlapping sets cannot be statistically compared, the persistence and slight numerical increase of the effect size in the specific ALT subgroup suggest that the association is not diluted or driven by nonspecific measures.

Nevertheless, the nonsignificant difference between measurement substantially shared variance among these constructs. Future research should aim to disentangle these components, perhaps by measuring explicit autistic traits, empathy, and social anxiety simultaneously, to determine

attentional atypicalities are driven by core social-cognitive impairments (as measured by explicit autistic traits or ToM; Johnson, 2014) or motivational factors (social anxiety/reward; Cañigueral & Hamilton, 2019; Edwards et al., 2015; Schurz et al., 2014). Until then, our results support the utility of the AQ and similar tools as sensitive indices for capturing individual differences in social attention.

Experimental parameters

Our analysis revealed no significant moderating effects of experimental parameters, suggesting that the negative association between ALT and social attention is robust across diverse experimental contexts. Specifically, neither face type nor expression significantly modulated the relationship, consistent with prior research noting the resilience of gaze cueing to variations in cue ecological validity and expression (McKay et al., 2021). This implies that individuals with higher ALT exhibit a generalized reduction in responsiveness to social cues (i.e., averted gaze faces), regardless of stimulus realism (schematic vs. realistic) or emotional content (e.g., fearful vs. neutral). However, wide confidence intervals observed in schematic and non-neutral conditions suggest that future studies with larger samples are needed to rule out subtle effects masked by low statistical power.

Regarding SOA, differences across SOA conditions were not statistically significant, although a "middle SOA" showed a numerical trend toward a stronger negative correlation. Variations in SOA are thought to reflect different stages of social attention processing (Shin et al., 2023): shorter SOAs typically trigger reflexive attentional orienting, whereas longer SOAs reflect voluntary and goal-driven processes (McKay et al., 2021; Yang et al., 2024). The consistent negative correlations across all conditions suggest that high ALT is associated with broad atypicalities in both reflexive and voluntary stages of social attention.

Similarly, task type (detection, localization, categorization) did not moderate the effect. Although localization and categorization tasks demand greater executive resources to reconcile cue-target congruency (Chacón-Candia et al., 2023; McKay et al., 2021), effect sizes in these conditions did not differ significantly from detection tasks. The absence of a modulation effect indicates that reduced social attention in high-ALT individuals is likely a pervasive characteristic, observable even in simple detection tasks, and is not solely dependent on the executive load of the paradigm.

Limitations and future directions

Several limitations should be acknowledged. First, by focusing exclusively on typically developing individuals, our analysis inherently restricts the range of ALT scores, particularly at the upper extreme. This range restriction likely attenuates the observed correlation strength relative to studies including clinical populations (Murza et al., 2016; Todorova et al., 2019). Future research should integrate both neurotypical and clinical samples to directly test the continuity of social attention mechanisms across the complete autism spectrum. Second, the majority of participants were recruited from Western (Europe/North America) and East Asian contexts. Cultural background may be a critical moderator, as social norms regarding eye contact vary significantly across cultures (e.g., the discouragement of direct gaze in some East Asian contexts; Cañigueral & Hamilton, 2019; Dalmaso et al., 2020; Gobel et al., 2017). Future studies should explore how culture-specific social norms interact with autistic traits to influence attentional orienting.

Third, while we examined basic experimental parameters (e.g., face type, expression), we could not sufficiently analyze the influence of higher-level social dimensions of the cue, such as race, gender (Chacón-Candia et al., 2023; Ricciardelli & Pintori, 2023), or social status (e.g., trustworthiness, reliability; Ristic & Capozzi, 2022). Although intervention studies suggest that manipulating social status modulates gaze cueing (Dalmaso et al., 2016; Hudson et al., 2012; Willemse et al., 2018), insufficient data prevented a meta-analytic test of this factor. Future work should systematically investigate how social identity cues interact with observer traits.

Finally, the reliance on static, computer-based gaze-cueing tasks limits the ecological validity of the findings. Decontextualized settings may fail to capture the dynamic, motivation-driven nature of real-world social attention (Pfeiffer et al., 2013). To better understand the mechanisms underlying social attention in ASD, future research should integrate ecologically valid paradigms, such as virtual reality (VR) or live interaction tasks, and incorporate multimodal social cues (Cañigueral & Hamilton, 2019; Leonhard et al., 2013). Such approaches would provide a more comprehensive understanding of how ALT influences social attention in complex, real-life environments.

Conclusion

In summary, this meta-analysis provides robust evidence for a weak but significant negative correlation between ALT and social attentional orienting in the general population. This finding supports the continuum hypothesis, suggesting that social attention difficulties are not confined to clinical ASD diagnoses but are dimensionally distributed across the broader population. Crucially, this association was most reliable when measured by explicit autistic traits instruments and was notably moderated by sample characteristics: the link was stronger in male-predominant samples and weaker in larger studies, highlighting the importance of statistical

power and sample composition. Conversely, the stability of the effect across diverse experimental parameters (e.g., face type, expression, SOA, task type) and measurement constructs points to a generalized, rather than context-specific, attenuation of social attention in high-ALT individuals. These insights clarify the mixed findings in the literature and underscore the need for future research to move beyond small-scale correlational designs. By adopting larger samples and more ecologically valid paradigms (e.g., dynamic interactions), researchers can better elucidate the precise mechanisms of social cognition across the autism spectrum, ultimately informing the development of targeted screening and intervention strategies.

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Code availability The code generated for this study are available on request to the corresponding author.

Declarations

Conflicts of interest The authors declare no conflicts of interest.

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Consent to participate Not applicable.

Consent for publication Not applicable.

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